

5.9 Relational Algebra

Relational algebra is a procedural query language that processes one or more relations to define another relation without changing original relations. The operands as well as the result are relations. The output of one operation can become the input of another operation to create nested expression in relational algebra. This property is known as **closure**.

5.9.1 Basic Operations of Relational Algebra

There are two categories of operations in relational algebra:

1. Unary Operations

The operations which involve only one relation are called **unary operations**. The following operations are the unary operations:

- Selection
- Projection

2. Binary Operations

The operations which involve pairs of relations are called **binary operations**. A binary operation uses two relations as input and produces a new relation as output. The following operations are the binary operations:

- Union
- Set Difference
- Cartesian Product

5.9.2 Selection Operation

The Selection operation is a unary operation. The selection operator is sigma σ . It acts like a filter on a relation. It returns only a certain number of tuples. It selects the tuples using a condition. The condition appears as subscript to σ . The resulting relation has the same degree as the original relation. However, the resulting relation may have fewer tuples than the original relation.

$\sigma_C(R)$ returns only those tuples in R that satisfy condition C

A condition C may consist of any combination of comparison or logical operators that operate on the attributes of R.

- Comparison operators: $= < > \geq \leq \neq$
- Logical operators: $\wedge \vee \neg$
- Use the Truth tables for logical expressions:

$\wedge$	T	F
T	T	F
F	F	F

$\vee$	T	F
T	T	T
F	F	F

$\neg$	T	F
	F	T

Examples

Assume a relation EMP has the following tuples:

Name	Office	Dept	Rank
Saleem	400	CS	Assistant
Junaid	220	Econ	Lecturer
Ghafoor	160	Econ	Assistant
Babar	420	CS	Associate
Saleem	500	Fin	Associate

Question

Select only those Employees who are in CS department:

$\sigma_{\text{Dept} = \text{'CS'}}(\text{EMP})$

Result

Name	Office	Dept	Rank
Saleem	400	CS	Assistant
Babar	420	CS	Associate

Question

Select only those Employees with last name Saleem who are assistant professors:

$\sigma_{\text{Name} = \text{'Saleem'}} \wedge \text{Rank} = \text{'Assistant'}}(\text{EMP})$

Result

Name	Office	Dept	Rank
Saleem	400	CS	Assistant

Question

Select only those Employees who are Assistant Professors or in Economics department:

$\sigma_{\text{Rank} = \text{'Assistant'} \vee \text{Dept} = \text{'Econ'}} (\text{EMP})$

Result

Name	Office	Dept	Rank
Saleem	400	CS	Assistant
Junaid	220	Econ	Lecturer
Ghafoor	160	Econ	Assistant

Question

Select only those Employees who are not in the CS department or Lecturer:

$\sigma_{\neg (\text{Rank} = \text{'Lecturer'} \vee \text{Dept} = \text{'CS'})} (\text{EMP})$

Result

Name	Office	Dept	Rank
Ghafoor	160	Econ	Assistant
Saleem	500	Fin	Associate

5.9.3 Projection Operator

Projection is also a unary operator. The Projection operator is π . It limits the attributes returned from the original relation. The resulting relation has the same number of tuples as the original relation. The degree of the resulting relation may be equal to or less than the original relation.

The general syntax is: $\pi_{\text{attributes}} R$

Where attributes is the list of attributes to be displayed and R is the relation.

Examples

Assume the same EMP relation used in previous examples.

Question

Display only the names and departments of the employees:

$\pi_{\text{name, dept}} (\text{EMP})$

Result

Name	Dept
Saleem	CS
Junaid	Econ
Ghafoor	Econ
Babar	CS
Saleem	Fin

Combining Selection and Projection

The selection and projection operators can be combined to perform both operations.

Question

Show the names of all employees working in the CS department:

$\pi_{\text{name}} (\sigma_{\text{Dept} = \text{'CS'}} (\text{EMP}))$

Result

Name
Saleem
Babar

Question

Show name and rank of those Employees who are not in CS department or Lecturers:

$\pi_{\text{name, rank}} (\sigma_{\neg (\text{Rank} = \text{'Lecturer'} \vee \text{Dept} = \text{'CS'})} (\text{EMP}))$

Result

Name	Rank
Ghafoor	Assistant
Saleem	Associate

5.9.4 Set Operations

The Selection and Projection operations extract information from one relation. The set operations are used to extract information from several relations. The relational algebra has set operations of Union, Set difference, Intersection and Cartesian product. These operations are also called **binary operations**.

5.9.4.1 Union

The union operation of two relations combines the tuples of both relations to produce a third relation. If two relations contain identical tuples, the duplicate tuples are eliminated. The notation for the union of two relations A and B is A UNION B. It is denoted by U.

If A is defined as $A = \{ a, b, c \}$ and set B is defined as $B = \{ a, c, 1, 2 \}$, then $C = A \cup B$ will return $C = \{ a, b, c, 1, 2 \}$.

The relations used in the union operation must have same number of attributes. The corresponding attributes must also come from same domain. Such relations are also called **union compatible relations**.

Union are commutative operations:

$$A \cup B = B \cup A$$

Example: $A \cup B$

Following is an example of union operation. Two relations A and B are combined together by using union operator.

Table A		
X	Y	Z
1	A	10
2	B	20
3	C	30

Table B		
X	Y	Z
1	A	10
4	D	40
5	E	50

A UNION B		
X	Y	Z
1	A	10
2	B	20
3	C	30
4	D	40
5	E	50

5.9.4.2 Difference

The difference operation works on two relations. It produces a third relation that contains the tuples that occur in the first relation but not in second. The difference operation can be performed on union compatible relations. The order of subtraction is significant.

The difference operator is not commutative. It means:

$$A - B \neq B - A$$

Example

Following is an example of difference operation. There are two relations A and B. The result of $A - B$ and $B - A$ are as follows:

Table A			Table B			$A - B$			$B - A$		
X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z
1	A	10	1	A	10	2	B	20	4	D	40
2	B	20	4	D	40	3	C	30	5	E	50
3	C	30	5	E	50						

Figure: The Difference operation

5.9.4.3 Intersection

The intersection operation works on two relations. It produces a third relation that only common tuples. Both relations must be union compatible. It is denoted by $\cap$.

Example

Following is an example of intersection operation. There are two relations A and B. The result of $A \cap B$ is as follows:

Table A			Table B			$A \cap B$		
X	Y	Z	X	Y	Z	X	Y	Z
1	A	10	1	A	10	1	A	10
2	B	20	4	D	40			
3	C	30	5	E	50			

Figure: The Intersection operation

5.9.4.4 Product

The product works on two relations. It concatenates every tuple in one relation with every tuple in second relation. It is also called **cartesian product** or **cross product**. The product of relation A with m tuples and relation B with n tuples is relation C with $m \times n$ tuples. The product is denoted as $A \times B$.

The Product needs not to be union compatible. It means that they can be of different degree. It is commutative and associative.

Table A			Table B			A X B					
X	Y	Z	X	Y	Z	X1	Y1	Z1	X2	Y2	Z2
1	A	10	1	A	10	1	A	10	1	A	10
2	B	20	4	D	40	1	A	10	4	D	40
3	C	30	5	E	50	1	A	10	5	E	50
						2	B	20	1	A	10
						2	B	20	4	D	40
						2	B	20	5	E	50
						3	C	30	1	A	10
						3	C	30	4	D	40
						3	C	30	5	E	50

Figure: The Product operation

5.9.4.5 Division

The division operator results in columns values in one table for which there are other matching column values corresponding to every row in another table.

A			B (Divisor)		Result
K	X	Y	X	Y	K
10	1101	A	1101	A	10
10	1201	B	1201	B	30
10	1301	C	1301	C	
20	1201	B			
30	1101	A			
30	1201	B			
30	1301	C			